

Phase Equilibria

Mrinal Kanti Roy

Lecturer (Chemistry)

Mymensingh Engineering College

Phase Rule

Gibbs phase rule may be stated as follows :

"In a heterogeneous system in equilibrium, the number of degrees of freedom plus the number of phases is equal to the number of components plus two".
Mathematically,

$$\mathbf{F + P = C + 2 \text{ or } F = C - P + 2}$$

Where,

F = number of degrees of freedom

C = number of components

P = number of phases

2 = additional variables of temperature and pressure besides the concentration variables.

Phase

“ The physically distinct, homogenous and mechanically separable part of a system are called phases”.

Examples (i) A gaseous mixture constitutes a single phase since gases are completely miscible. Air is a mixture of N_2 , O_2 , CO_2 , water vapour etc. Which constitute a single phase.

(ii) Two or more liquids which are miscible with one another constitute a single phase as there is no bounding surfaces separating the different liquids. e.g., water and alcohol, benzene & chloroform constitute one phase system.

(iii) A system consisting of a liquid in equilibrium with its vapour constitute a two phase system

Component

The number of component of a system at equilibrium is defined as the minimum number of independently variable constituents which are required to express the composition of each phase in the system.

In a chemically reactive system, the number of components is given by

$$C = N - E$$

where C = components.

N = Number of chemical species

E = Number of independent equations relating the concentrations of the N species.

Each independent chemical equilibrium involving the constituents counts as one equation. The condition that a solution be electrically neutral also counts as one equation if ions are considered as constituents.

Examples

- (i) Sulphur system:** Consists of four phases namely monoclinic sulphur, rhombic sulphur, liquid sulphur and sulphur vapour. The composition of each phase of the system can be expressed in terms of sulphur only, so, it is a one component system.
- (ii) Water system:** It is a one component system because the composition of each of the three phases present can be expressed as H_2O .
- (iii) Na_2SO_4 + water system:** Certain salts are capable of existing as hydrates with different number of water molecules of crystallization. These hydrates correspond to different solids and hence to different phases. The system is a two component, because the composition of each phase of the hydrates is completely described in terms of the anhydrous salt and water alone. e.g., Na_2SO_4 + water

Degrees of Freedom

The degree of freedom or variance of a system is defined as the minimum number of variable factors such as temperature, pressure and concentration which should be arbitrarily fixed in order to define the system completely.

Examples

(i) For a given sample of any gas $PV = nRT$. Any two of the three variables P , V , T define the system completely. Hence the system is bivariant or it has two degrees of freedom.

(ii) A gaseous mixture say N_2 and O_2 gases (mixed 50% each), is completely defined when three variables temperature, pressure and concentration are specified. Thus, the degrees of freedom is three or the system is trivariant.

Thermodynamic Derivation of Phase Rule

Gibbs rule relies greatly on the Gibbs-Duhem equation, which is a fundamental basis for thermodynamics. The Gibbs-Duhem equation clarifies the relationship between pressure (P), temperature (T) and potential for chemical components (μ). This equation is as follows $dG = Vdp - Sdt + \sum N_i d\mu_i$

However, a simpler way to derive the Phase rule is to understand that the composition of each phase is defined as P (C-1). Thus, the total number of variables is equal to P (C-1) + 2

Number of equilibrium for each component's each phase is P-1

Now for C number of components, the number of equilibrium = P (C-1)

Therefore, total number of equilibria $E = C (P-1)$

Now,

$$F = \{P (C-1) + 2\} - \{C (P-1)\}$$

$$F = \{CP - P + 2 - CP + C\}$$

$$F = C - P + 2$$

The Water System

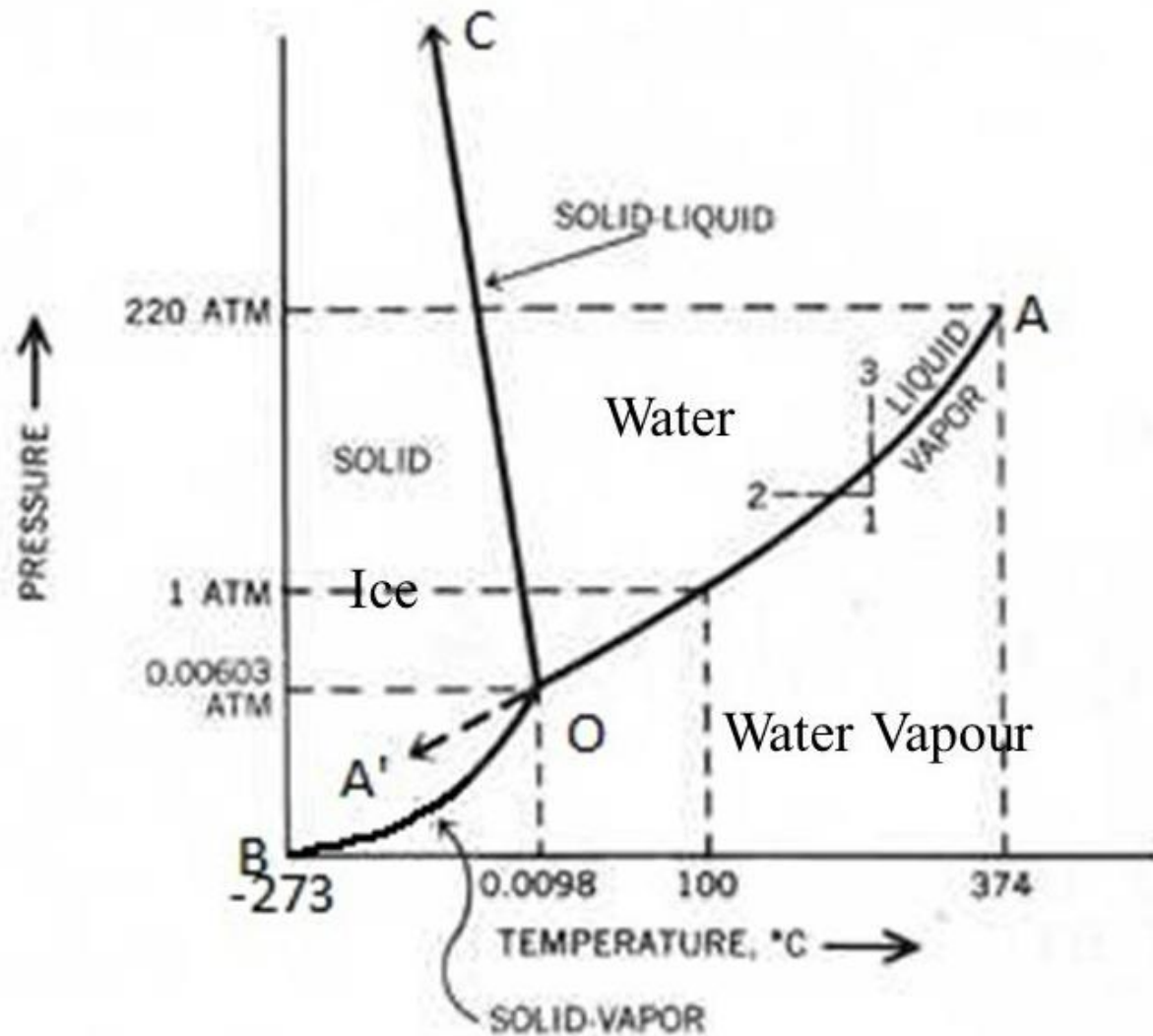
It is a one component system. Water exists in three possible phases. ice (solid) , water (liquid), and vapour (gas). These three single phases may form four possible equilibria.

(i) Solid $\rightleftharpoons$ Liquid

(ii) Liquid $\rightleftharpoons$ Vapour

(iii) Solid $\rightleftharpoons$ Vapour

(iv) Solid $\rightleftharpoons$ Liquid $\rightleftharpoons$ Vapour



Water System

The phase diagram consists of :

- (i) **Stable curves:** three OB, OA and OC
- (ii) **Metastable curve:** one OA'
- (iii) **Areas:** three AOB, COB and AOC
- (iv) **Triple point:** One O

OA : It is known as vapour pressure curve of water. The curve OA starts from point O i.e., freezing point of water, 0.0098°C under 4.579 mm of Hg pressure and ends at A, the critical temperature (374°C at 218 atm.). Above critical temp. on the vapour phase exists whatever may be the value of pressure. The vapour pressure of water increases with increase in temperature

The rate of increase of its vapour pressure with temperature is relatively higher at higher temperatures and therefore the curve OA slants upwards and slopes away from the temperature axis.

From phase rule, $F = C - P + 2 = 1 - 2 + 2 = 1$

The water vapour system is univariant

OB : It is the sublimation curve of ice. Along this curve, solid ice is in equilibrium with its vapour. This curve is not the prolongation of curve A but falls off more steeply. Curve OB starts from the temperature 0.0098°C above which solid water i.e., ice cannot exist. The curve terminates at B i.e., absolute zero (-273°C). At this temperature, no vapour can exist and, therefore, only ice is left. But on other points of the curve OB, ice is in equilibrium with vapour.

Hence, there are two phases. According to phase rule,

$$F = C - P + 2 = 1 - 2 + 2 = 1$$

Thus, the system is univariant.

This means that for each temperature; there may be one pressure and for each pressure there may be one temperature

OC : This curve is the melting point curve or fusion curve of ice. Along this curve two phases, ice and water are in equilibrium. The inclination of OC line towards the pressure axis indicates that the melting point of ice is slightly lowered by increase of pressure. (According to Le Chatelier's principle the increase in pressure causes the water - ice equilibrium to shift in such a direction that there is a decrease in volume.) As the melting point of ice is accompanied by decrease in volume, it should be lowered by the increase of pressure. The curve OC starts from point O but there is no limit for this curve. It goes upto a point corresponding to 2000 atm. and -20°C.

According to phase rule,

$$F = C - P + 2 = 1 - 2 + 2 = 1$$

Thus, the system is univariant.

This means that for any given pressure, melting point must have one fixed value.

OA' : It is a **metastable curve** shown in continuation of AO. When water is cooled below its freezing point (when it is not vigorously stirred) without separation of ice, the water is said to be super cooled. The vapour pressure curve of liquid water AO extends below O as shown by the dotted curve OA'. Along curve OA' liquid water coexists with vapour and vapour pressures are different than over the solid. This equilibrium is called metastable equilibrium as slight disturbance brings it to the stable region OB of the phase diagram.

(iii) Areas: The areas give the conditions of temperature and pressure under which single phase - ice (solid), water (liquid) and vapour (gas) can exist. It is necessary to specify both temperature and pressure to define a system within this area. In the area BOC, AOC and AOB exists, ice (solid), water (liquid) and vapour (gas) respectively. In these areas, the degrees of freedom for the system is two or they are bivariant

(iv) The Triple point O: The point O at which the curves AO, BO and CO meet is called the triple point. At this point all the three phases viz , ice, water and vapour co-exist. Thus, $P = 3$. According to phase rule, at triple point O,

$$F = C - P + 2,$$

$$= 1 - 3 + 2,$$

$$F = 0$$